Diet Optimization Project Report

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Abstract

This project focuses on balancing a person's daily nutritional intake by optimizing their fourth meal of the day. Using data from the Anuvaad Indian Nutrition Databank, we designed a mathematical model to calculate the exact nutrients required so as to ensure a healthy daily diet. We are using Kullback-Leibler (KL) Divergence to measure the divergence between the current nutrient proportions and the ideal nutrition targets(i.e. Recommended Dietary Allowance(RDA)). By considering the medical limits(i.e. Estimated Average Requirement(EAR) and Tolerable Upper Limit(TUL)) to prevent both deficiency and excess of nutrients, our model provides a personalized nutrient target for the fourth meal.

Introduction

Eating a balanced diet is very important for good health, but it is hard to track every single nutrient consumed. Usually, people eat breakfast, lunch, and a snack without calculating their exact macro and micro nutrients' intake. Because of this, our daily diet often misses the required targets.

This paper attempts to solve this issue by focusing on the fourth meal of the day. If we know what nutrients had already been consumed in the first three meals, we can calculate the remaining amount/proportion of nutrients from the required/referenced level. Using mathematical optimization, we have designed the ideal fourth meal to ensure a balanced diet.

Problem Statement

Very often, the person is usually unaware whether the previous meals consumed have a specific nutrient more (or less) than what is tolerable (or required) therefore, it is necessary to keep their daily diet healthy with respect to their gender, age group, and physical activity. This problem aims to specify the optimal nutrition level for the fourth meal of the day.

Plan

Given the three meals of the day, we plan to find the optimal nutrient proportions for the fourth meal such that the entire nutrient intake is as close to the RDA as possible. For this, we first consider the given three meals, which were randomly taken from the dataset. Then, we took the RDA as target nutrient intake profile for 9 chosen nutrients. Now, to choose the optimal nutrient proportion for the fourth meal, we formulated an objective function alongside possible constraints upon intake of nutrients, which are EAR and TUL, using Kullback-Leibler (KL) Divergence. The reason to consider KL-Divergence is its sensitivity to nutrients that are missing ($p_i > 0$) but required in the diet ($r_i > 0$). We then solved for the optimal value of objective function using R and Python.

Dataset

We took the dataset from Anuvaad Indian Nutrition Databank. It lists amounts of nearly 40 nutrients and minerals for 1014 dishes (mostly Indian; few are from US and UK). The data for each item is present in two forms: per 100g and unit serving. The data collected by Anuvaad is sourced from several official journals e.g., ICMR, CoFID etc., as well as blog posts.

The data for ideal nutrient intake profile and constraints i.e. Recommended Dietary Allowance (RDA), Estimated Average Requirement(EAR) and Tolerable Upper Limit(TUL) was taken from the website of Department of Health Research, Ministry of Health and Family Welfare.

Data Cleaning

The original Nutrition Databank only consisted variables like, "food_name", "food_code", "primarysource", and the other eighty for the two types of nutritional info. So, to go in line with the problem statement, the variables, category (drinks, main, dessert etc.), breakfast, lunch etc., were to be added. Moreover, it had to be ensured that the units for all the values were same. Techniques like, "pattern matching and replacement" and "split-apply-combine" have been used for the former and the later respectively.The grepl function has been used for pattern matching; it takes missing values in x as not matching a non-missing pattern. A little mismatching in the categories might have happend as we relied fully on pattern matching using the keywords in the "food_name" column. The ordinary apply function and vectorized features in R have been extensively exploited. For data cleaning, only the base package has been used. All the nutritional values were eventually converted into grams (from micrograms, milligrams, IU).

Despite the availability of data on 23 body types and 40 nutrients, only 6 body types and 9 nutrients have been considered in the optimization problem. The bold drawback was the NA values in the dataset. The TUL for protein is in percentage and the RDA for energy is not available in the reference table; thence they were dropped. Further, to ensure that the arguments in the objective function i.e., KL Divergence, were probability distributions we had to convert the RDA and the nutritional values into proportions (discrete distributions).

Analysis

In our analysis, KL divergence acts as a precision tool for nutrition balance by treating the remaining RDA requirements as a probabity distribution. The main advantage of using KL divergence is that it focuses on the relative deficiency of each nutrient. By minimizing the divergence between the nutrients of the fourth meal and the Remaining RDA requirements(R), we ensure that the overall nutrient intake is as close to the RDA for the day.

Before applying the diverngence formula, we must define the few mathematical notations as followings:

- Nutrient set: We have overall 9 nutrients to compare, let $i \in \{1, 2, \ldots, 9\}$ represent the set of nutrients in the Food Items.
- Food Items set(F): In our dataset we have 1014 different food items. Let n be the number of items in the consumed meals, and $j \in \{1, 2, \ldots, x\}$ represent the set of food items for the fourth meal (x is the total options for the dinner).
- TUL Proportion Vector(M): This contains the TUL proportions of all 9 nutrients, $M_i = \frac{TUL_i}{\sum TUL_i}$.
- EAR Proportion Vector(N): This contains the EAR proportions of all 9 nutrients, $N_i = \frac{EAR_i}{\sum EAR_i}$.
- Consumption Vector(C): This contains the values of sum of nutrients consumed in the first three meals. Here, c_i = sum of ith nutrient consumed in the three meals.
- RDA Vector(RDA): RDA values of all the 9 nutrients, RDA_i = the RDA value of ith nutrient.
- Target Gap Vector(T): This shows the remaining RDA requirements after the first three meals.Here, $t_i = |RDA_i - c_i|$ (remaining RDA requirement for i^{th} nutrient).

Then we define the distributions:
Since KL Divergence is defined over probability spaces, we are going to normalize the Target Gap Vector(T) and treat nutrient values as probability,

- Target Distribution(R): We normalize the target gap such that $r_i = \frac{t_i}{\sum t_i}$, this represents the distrbution of the target gap for the day.
- True Distribution of jth Food item(X_j): For the jth food item, the ith nutrient is represented as $x_{ji} \in \{\text{x}_{j1}, x_{j2}, \ldots, x_{j9}\}$ and this represents the distribution of the jth food item's proportions of nutrients.

Mathematical Formalization

With our nutritional data transformed into probability distributions X_j and R, we define the objective functions:
Objective function for finding the Optimal Food Item (without any constraints)

$$\hat{F} = \arg\min_{F_j} \sum_{i=1}^{9} x_{ji} \log\left(\frac{x_{ji}}{r_i}\right)$$

Here, $\hat{F}$ is the Optimal food item for the objective function $\arg\min_{F_j} \sum_{i=1}^{9} x_{ji} \log\left(\frac{x_{ji}}{r_i}\right)$. This objective function helps to identify the items in the food set that should be optimal to consume for keeping the divergence from the RDA values as minimum as possibe. **Objective function for finding the Optimal Nutrient Proportions**

$$\begin{aligned} \textit{minimize} \quad & \sum_{i=1}^{9} x_i \log\left(\frac{x_i}{r_i}\right) \\ \text{subject to} \quad & x_i \leq M_i, \quad i = 1, \ldots, 9 \\ & x_i \geq N_i, \quad i = 1, \ldots, 9 \\ & \sum_{i=1}^{9} x_i = 1 \end{aligned}$$

Where $x \in \mathbb{R}^9$ is the optimal solution. With the given constraints the optimal solution(x) for the objective function gives us the optimal nutrient proportions to counsume in the fourth meal.

Computation in R and Python

The random component has been introduced by taking a random sample from the available 1013 data points (the 707th entry- "Jam filling", had 0 g of each of the selected 9 nutrients). The function, sample_n, from the library dplyr has been used for so with desired sample sizes for the given three meals (say, breakfast, dinner and snacks). As the optimal solution, the nutritional proportions ensuring that the entire (four times) meal consumption is as close to the RDA proportions as possible will be returned.

For the unconstrained problem, the data entry corresponding to the minimum objective value will be returned along with the name of the food item. Whereas, in the constrained problem, only the optimum nutritional proportions will be returned suggesting an ideal food item.

The unconstrained part has been easily formulated with a user made function in R. Due to multiple dependencies of the library CVXR i.e., rust, fortran, matrix etc., we preferred switching to CVXPY (in Python) for the constrained part of the problem. Recall, the true and the target distributions are X_j (the nine nutrition values of jth food item) and r_i (the Target Gap Vector). The inbuilt function kl_div, in CVXPY, we used has the formula:

$$\textit{minimize} \quad \sum_{i=1}^{9} [x_i \log\left(\frac{x_i}{r_i}\right) + x_i - r_i]$$

Since x_i and r_i are two different proportions for all $i \in \{1, 2, \ldots, 9\}$, they sum upto one individually hence, cancelling each other out and preserving the formula in the previous section.

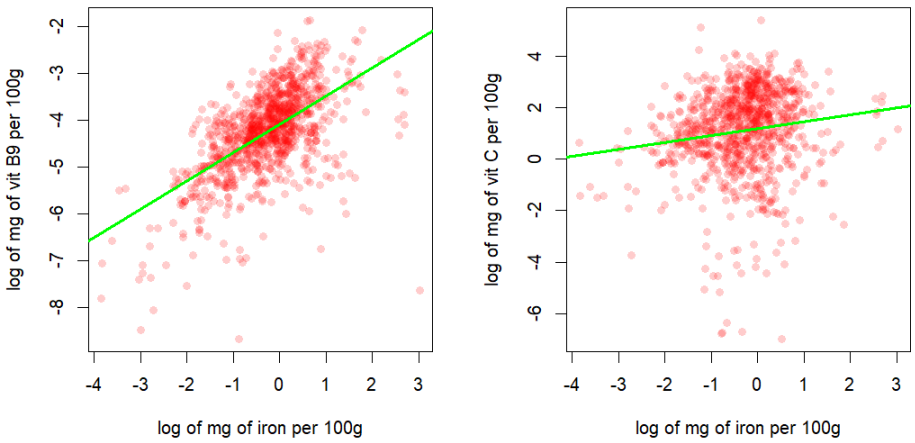

Figure 1: B9 is more correlated with Iron than vit C

Limitations

While our mathematical model calculates the optimal value, the process has a few limitations:

- In the KL-Divergence formula, calculating the logarithm when a nutrient's proportion (p_{ij}) is exactly zero, the program returns NaN (since log(0) is undefined). Perhaps, another objective function could've resolved it.
- We assumed that nutrient weights (x_{ij}) add up linearly. But biologically, this is not entirely true because some minerals stop others from being absorbed, which our equations do not capture.
- KL-Divergence is excellent for making the *ratios* of nutrients match the ideal RDA perfectly. However, having the correct ratios does not guarantee that the absolute intake (such as total calories or grams consumed) is sufficient to meet energy needs. In our case, since the dataset does not include calorie information, the optimization is performed only over nutrients and minerals, and therefore does not ensure that the fourth meal provides adequate energy. This can lead to practical issues: if the meal contains too few calories, it may fail to supply sufficient energy to the body, while excessive calories may result in overconsumption, where the surplus energy is stored as body fat, potentially leading to weight gain and, over time, overweight or obesity.
- The model does not account for individual variability such as metabolic differences, allergies, food intolerances, or dietary restrictions (e.g., vegetarianism, lactose intolerance), which significantly affect real-world meal planning.
- The analysis assumes perfect and accurate nutrient measurements from the dataset. Any inaccuracies, rounding errors, or inconsistencies in the nutri-

tional data directly affect the optimization results and may lead to misleading recommendations.

- The consumed and the suggested optimal nutrients' amount are for 100 g of every item. But very often, the person eats significantly more or less than this amount.

Conclusion

To summarize, we treated daily nutritional balance as a mathematical distribution matching problem. By taking the first three meals as fixed constants, we were left with a nutritional gap. We set up KL-Divergence as our objective function to minimize the distance between the fourth meal's nutrient proportions (p) and our ideal RDA target distribution (r). To ensure the results were medically safe, we bounded this objective using 19 constraints based on EAR (to prevent deficiency) and TUL (to prevent excess).

Mathematically, the model works perfectly to output an optimal final meal. However, when interpreting this process for real-life diets, we must consider its limitations. The optimization might point to a mathematical coordinate that does not exist in our dataset (an empty feasible region), or it might give us a perfect nutrient ratio that is physically too small or large to actually eat. Additionally, since calorie information is not included, the model does not ensure that the meal provides adequate energy.

In conclusion, this project shows how mathematical methods can be used to improve daily diet planning. While the model works well in theory to balance nutrients, applying it to real meals is still challenging. In the future, adding factors like calorie intake and making sure the suggested meals are practical to eat can make the model more useful in real life.

Supplementary Material

Proof of Convexity of the Optimization Problem

We use the notations,

$$
\begin{aligned}
& f_0(x) = \sum_{i=1}^{9} x_i \log\left(\frac{x_i}{r_i}\right) \\
\text{subject to} \quad & f_1(x) = x - M \preceq 0, \quad i = 1, \ldots, 9 \\
& f_2(x) = N - x \preceq 0, \quad i = 1, \ldots, 9 \\
& g(x) = \sum_{i=1}^{9} x_i - 1 = 0
\end{aligned}
$$

And the Domain of the problem is $x \in \mathbb{R}^9$ which is *vacuously* convex.

Proof for Convexity of Objective function

For $x \in \mathbb{R}^9$, compute the first derivatives of f:

$$\begin{aligned}\frac{\partial f}{\partial x_i} &= \frac{\partial}{\partial x_i}\big(x_i \ln x_i - x_i \ln r_i\big) \\ &= \ln x_i + 1 - \ln r_i,\end{aligned}$$

Then we compute the Second partial derivatives (Hessian matrix),

$$\frac{\partial^2 f}{\partial x_i^2} = \frac{\partial}{\partial x_i}\big(\ln x_i + 1 - \ln r_i\big) = \frac{1}{x_i},$$

$$\frac{\partial^2 f}{\partial x_i \, \partial x_j} = \frac{\partial}{\partial x_j}\big(\ln x_i + 1 - \ln r_i\big) = 0$$

Hence the Hessian matrix of f is

$$H_f = \begin{pmatrix} \frac{\partial^2 f_0}{\partial x_1^2} & \frac{\partial^2 f_0}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f_0}{\partial x_1 \partial x_9} \\ \frac{\partial^2 f_0}{\partial x_2 \partial x_1} & \frac{\partial^2 f_0}{\partial x_2^2} & \cdots & \frac{\partial^2 f_0}{\partial x_2 \partial x_9} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f_0}{\partial x_9 \partial x_1} & \frac{\partial^2 f_0}{\partial x_9 \partial x_2} & \cdots & \frac{\partial^2 f_0}{\partial x_9^2} \end{pmatrix}$$

After substituting the derivative values,

$$H_f = \begin{pmatrix} \frac{1}{x_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{x_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{x_9} \end{pmatrix}$$

And according to the Sylvester's criterion we can say that the Hessian matrix H_f is Positive Semi Definite,

$$H_f \succeq 0$$

Therefore, f_0 is a convex function.

Proof for Convexity of Inequality Constraints

Here, f_1 and f_2 are similar to $Ax + b$ which is a convex function. Therefore, the inequality constraints are convex.

Proof for Affine of Equality Constraints

The function is $g(x)$ is affine, since it represents a hyperplane.

Team Contributions :

There were equal contributions in value by all three team members. The members researched their parts and wrote the latex code as well.

- Omhari Indoria (BSDDH2516) took initiative for the PROBLEM, PLAN and DATASET (partial) sections.
- Valluri Leela Sai Krishna (BSDDH2522) worked on the DATASET (partial) and ANALYSIS sections.
- Pawan Kumar (BSDDH2517) handled the CONCLUSION section and led the team in the writing the R AND PYTHON CODE for the optimisation model.

We divided different sections of the report among ourselves, but the overall research and the R code for the model were completed collaboratively with equal effort and dedication from all members.